

PERSONAL DATA

PLACE AND DATE OF BIRTH: Kolhapur, Maharashtra, India | July 1994
NATIONALITY AND GENDER: Indian | Female
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POST-DOCTORAL POSITIONS

TENURE: July 2026 - present
ADVISOR/HOST: [Dr. Jiaxin Han](#)
INSTITUTE: Shanghai Jiao Tong University, Shanghai

TENURE: APR 2025 - MAR 2026
ADVISOR/HOST: [Dr. Nishikanta Khandai](#)
INSTITUTE: National Institute of Science Education and Research (NISER), Bhubaneswar

PHD

TENURE: AUG 2018 - DEC 2024
TOPIC: Aspects of gravitational clustering and structure formation in the Universe
ADVISOR: [Prof. Jasjeet Singh Bagla](#)
INSTITUTE: Indian Institute of Science Education and Research (IISER), Mohali

EDUCATION

JULY 2017 Master of Science in PHYSICS,
Department of Physics, Savitribai Phule Pune University, Pune
Major: Astronomy and Astrophysics | CGPA: 8.27/10

JULY 2015 Bachelor of Science in PHYSICS
DBJ College, Chiplun, **Mumbai University** | CGPA: 7/7
Secured third rank at the University level

APR 2012 Higher Secondary Certificate
DBJ College, Chiplun, **Maharashtra State Board** | PERCENTAGE: 88.17%

APR 2010 Secondary School Certificate
Alore High school, Chiplun, **Maharashtra State Board** | PERCENTAGE: 94.18%

RESEARCH INTEREST

- Gravitational clustering and large-scale structure formation through cosmological simulations and analytical modeling, linking theory with survey observations.
- Dark matter halos, galaxy formation, and baryonic processes

PUBLICATIONS [NASA-ADS](#)

2022 : Swati Gavas, Jasjeet Bagla, Nishikanta Khandai, Girish Kulkarni
Halo mass function in scale invariant models,
[MNRAS, Volume 521, Issue 4, June 2023, Pages 5960–5971](#)

- 2024 : Swati Gavas, Jasjeet Bagla, Nishikanta Khandai
Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering,
[Phys. Rev. D 111, 043516 – 10 February, 2025](#)
- Jasjeet Bagla, Swati Gavas
On the origin of transient features in cosmological N-Body Simulations,
[JOAA, Volume 46, article number 33, 16 June 2025](#)
- 2025 : Dipayan Mukherjee, Harkirat Singh Sahota, Swati Gavas
A dynamical systems perspective on the thermodynamics of late-time cosmology
[Fortschritte der Physik, VL:74, IS:4, SN:0015-8208](#)
- 2026 : Ayan Nanda, Nishikanta Khandai, Jasjeet Singh Bagla, Swati Gavas
Universality of Halo Shape and its Morphological Evolution across Cosmic Time.
[astro.ph-CO](#)
- [In preparation]
Swati Gavas, Ayan Nanda, Nishikanta Khandai
Finite box size effect on halo scales
- [In preparation]
Ayan Nanda, Nishikanta Khandai, Swati Gavas
Multipole Characterization of Triaxial Dark Matter Halos:
A Spherical Harmonic Approach

CONFERENCES, SCHOOLS AND WORKSHOPS

- 29-2 OCT 2025: (presented talk)
Challenges and Innovations in Computational Astrophysics VI
Held at Indian Institute of Science Education and Research (IISER), Mohali
Talk title: Numerical Artifacts of Restricted Power Spectrum in cosmological N-Body Simulations
- 11 Nov 2024: (presented talk)
Computational and Observational Cosmology Group at the Observatório Nacional - ON(RJ/Brazil)
Talk title: Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering
- 14-25 OCT 2024: **3rd IAGRG School on Gravitation and Cosmology**
Held at International Centre for Theoretical Sciences (ICTS), Bengaluru, India
- 1-2 JULY 2024: (presented talk)
Séminaire Univers Institut d'Astrophysique de Paris (IAP)
Visit to Prof. Stephane Colombi at Institut d'Astrophysique de Paris (IAP), France
Talk title: Halo mass function in scale-invariant models
- 17-28 JUNE 2024: (presented talk)
Summer School on Cosmology
Held at International Centre for Theoretical Physics (ICTP), Trieste, Italy
Talk title: Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering

- 1-4 FEB 2024: (presented talk)
42nd Meeting of Astronomical Society of India
 Held at IISc, ISRO and JNP Bengaluru, India
 Talk title: Dispersion in the Hubble-Lemaître constant measurements from gravitational clustering
- 6-9 DEC 2023: (presented talk)
10th International Conference on Gravitation and Cosmology
 Held at IIT Guwahati, India
 Talk title: A Local Perspective on Hubble Tension from Cosmological N-body Simulations
- 7-9 NOV 2023: (presented talk)
Challenges and Innovations in Computational Astrophysics V
 Virtual Meeting
 Talk title: A Local Perspective on Hubble Tension from Cosmological N-body Simulations
- 24-28 OCT 2022: (presented talk)
The 10th KIAS Workshop on Cosmology and Structure Formation
 Held at KIAS, South Korea
 Talk title : Halo mass function in scale invariant models
- 2-6 NOV 2020: (presented poster talk)
The 9th KIAS Workshop on Cosmology and Structure Formation
 Held at KIAS, South Korea
 Poster title : Fractal dimension: Scale of homogeneity
- 13-17 FEB 2020: (presented poster)
38th Meeting of Astronomical Society of India
 Held at IISER Tirupati, India
 Poster title : Non-universality of halo mass function using power law model
- 27-31 JAN 2020: **School on Observing The First Billion Years of the Universe Using Next Generation Telescopes**
 Five days school held at IIT Indore, India
- 10-13 DEC 2019: (presented poster and volunteered)
9th International Conference on Gravitation and Cosmology
 Held at IISER Mohali, India
 Poster title : Non-universality of halo mass function using power law model

FELLOWSHIPS, SCHOLARSHIPS & AWARDS

- ANRF-ITS 2024: International travel grant to attend school in Italy
- INSA-INYAS SCIART 2021: International SciArt Image Competition 2021
 Third prize in simulation category for first entry
 Certificate of appreciation for second entry
- UGC-SET 2018: For lectureship in physics
- INSPIRE FELLOWSHIP 2017: Fellowship for Doctorate program in physics
- GATE PHYSICS 2018: All India Rank- 484
- JAM PHYSICS 2015: All India Rank- 460
- INSPIRE SHE: INSPIRE Scholarship for Higher Education
 Tenure: 2012-2017

OTHER RESEARCH EXPERIENCE

- JAN-MAY 2017 **Masters project at IUCAA, Pune, India**
N-body simulations to study large scale structure of the universe and study halo properties.
Supervisor : Dr. Aseem Paranjape, IUCAA, Pune, India.
- JUN-JULY 2016 **Summer School Project at IPR, Gandhinagar, India.**
Time dependent electron-ion collision frequency in a strong laser field using non-Maxwellian electron velocity distribution function
Supervisor : Dr. Mrityunjay Kundu, IPR, Gandhinagar, India.

TEACHING & MENTORING

- JAN-PRESENT 2025: Co-mentoring a PhD student as a part of collaboration
- AUG-DEC 2022: PHY111 Physics lab Mechanics (IISER Mohali)
- JAN-MAY 2021: IDC201 Introduction to astronomy (IISER Mohali)
- JAN-MAY 2020: PHY212 Modern Physics Lab (IISER Mohali)
- AUG-DEC 2019: PHY411 Nuclear Physics Lab (IISER Mohali)

TECHNICAL SKILLS

- HIGH PERFORMANCE COMPUTING : [Kalinga\(NISER\)](#), [astro \(NISER\)](#), [HPC-IISERM](#)
- PROGRAMMING LANGUAGES : Python, Julia, Fortran90, C/ C++.
- OPERATING SYSTEMS : Linux based systems
- TYPESETTING SOFTWARE : $\text{\LaTeX}$

LANGUAGES

- MOTHER TONGUE: Marathi
- FLUENT: English, Hindi

OTHERS

- MAY 2017: Video on 'zero shadow day' shown in outrageous acts of sciences (UK) / you have been warned (USA), a Discovery channel show.
- DEC 2009: **National Means Cum-Merit Scholarship Scheme (NMMSS)**
Tenure: 2009-2012